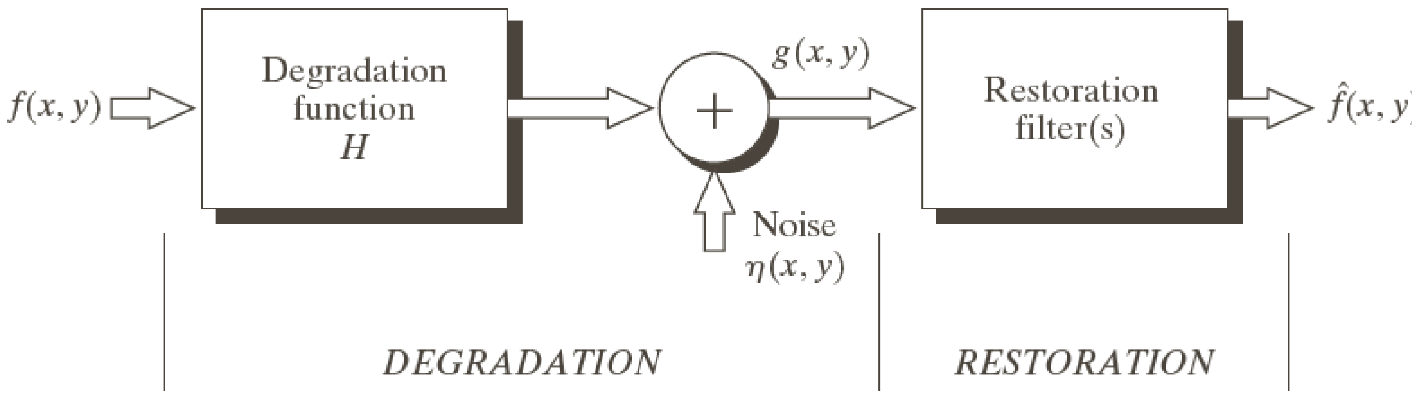
Chapter 5
Image Restoration

(Overview)

Degradation Model

$g(x,y)=H[f(x,y)]+n(x,y)$

FIGURE 5.1 A model of the image degradation/ restoration process.



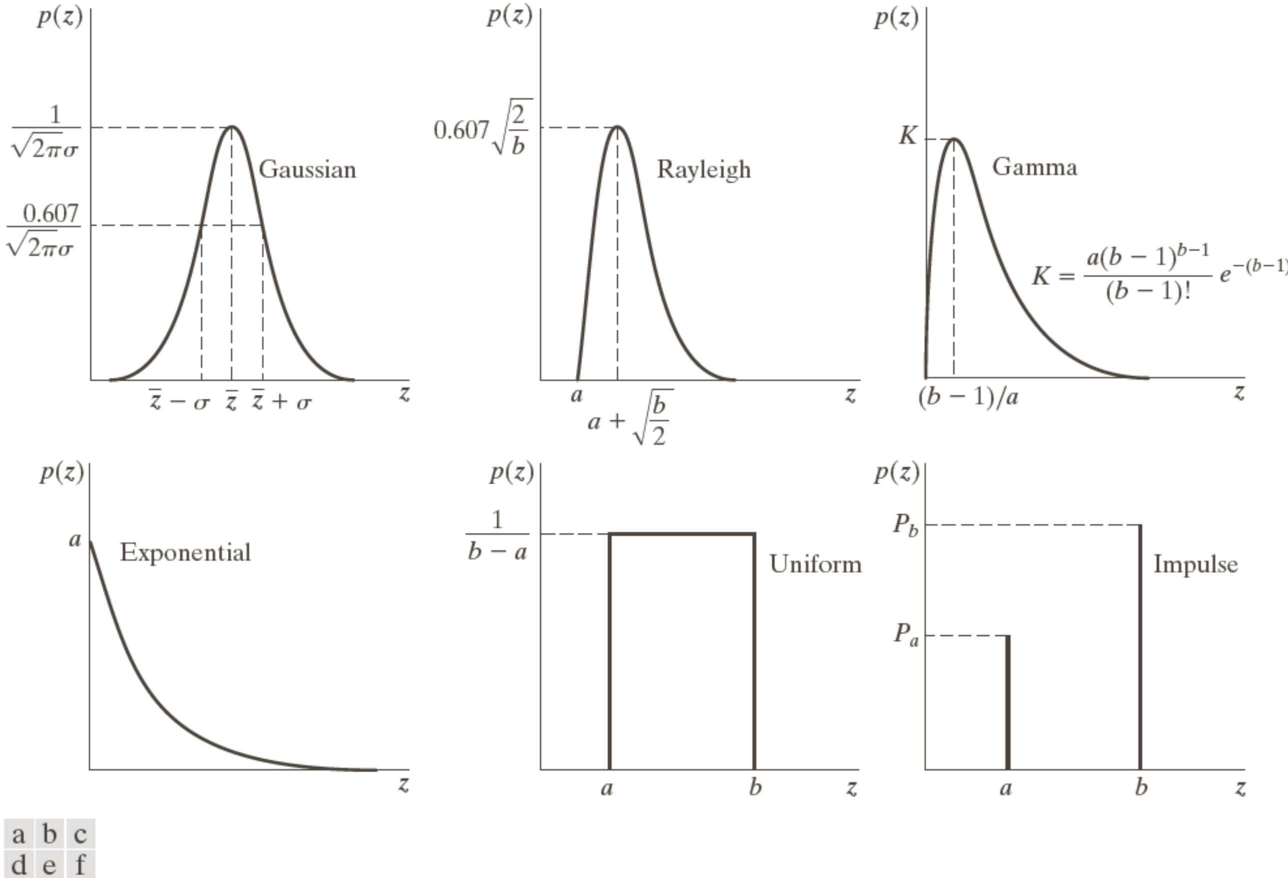


FIGURE 5.2 Some important probability density functions.

FIGURE 5.3 Test pattern used to illustrate the characteristics of the noise PDFs shown in Fig. 5.2.

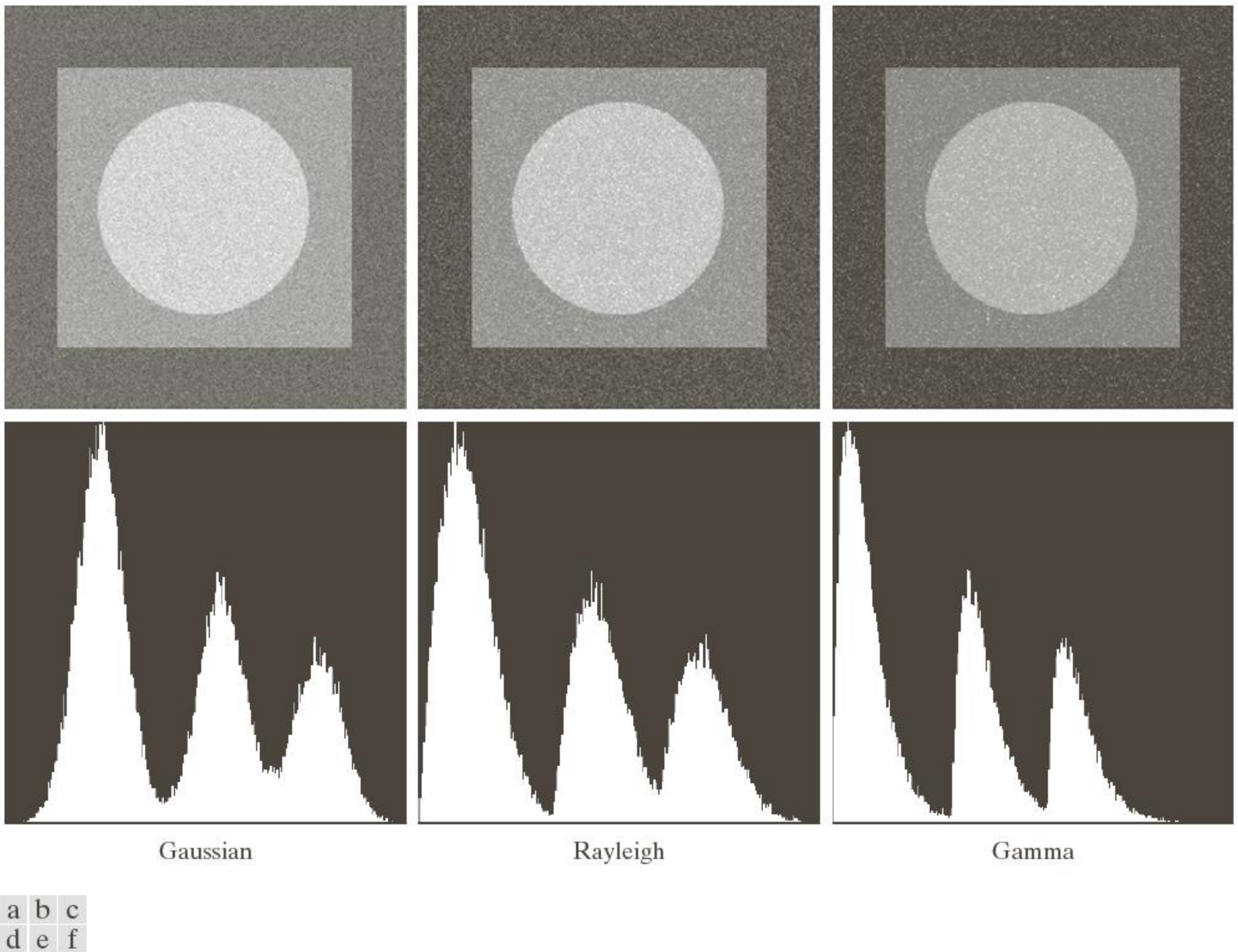


FIGURE 5.4 Images and histograms resulting from adding Gaussian, Rayleigh, and gamma noise to the image in Fig. 5.3.

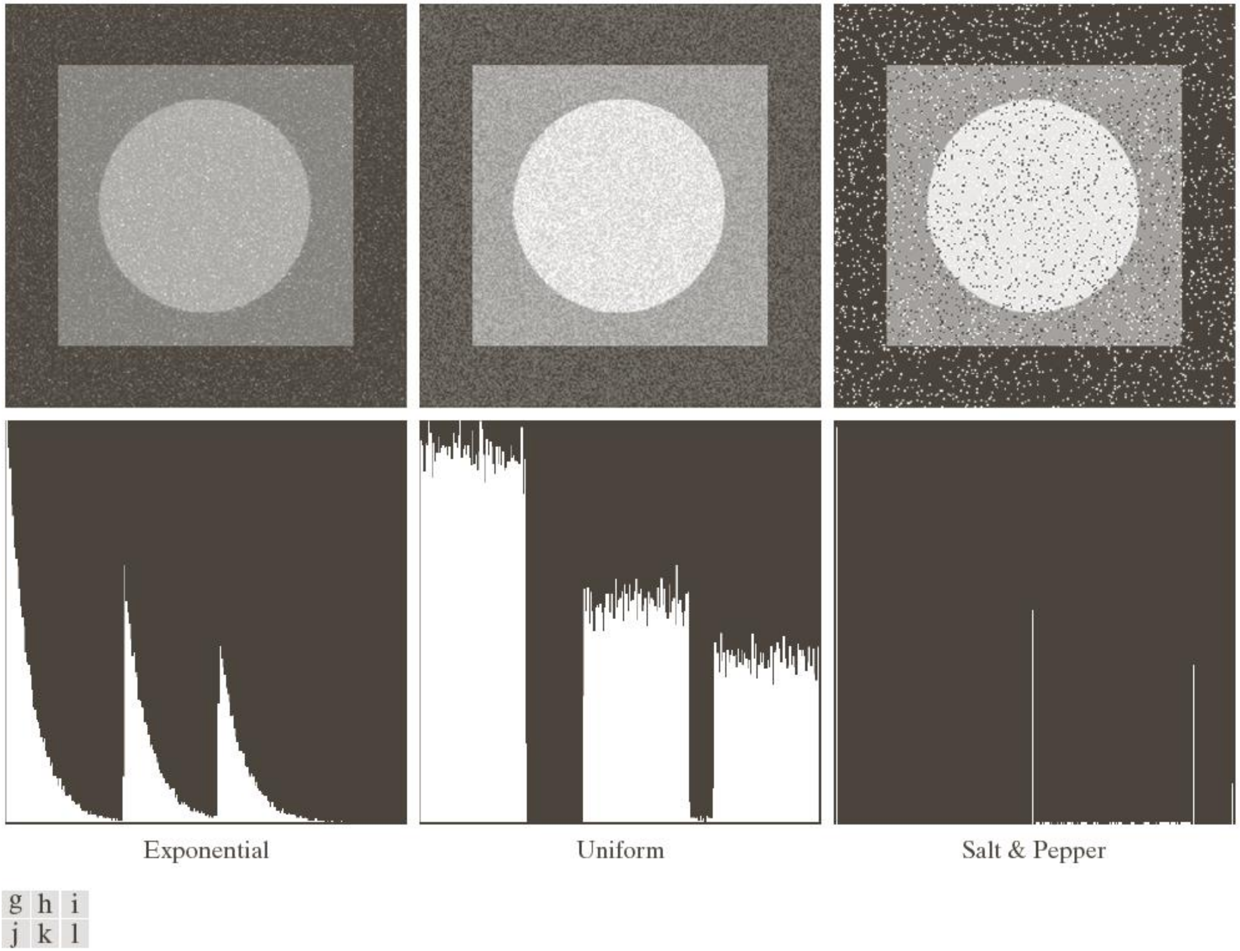
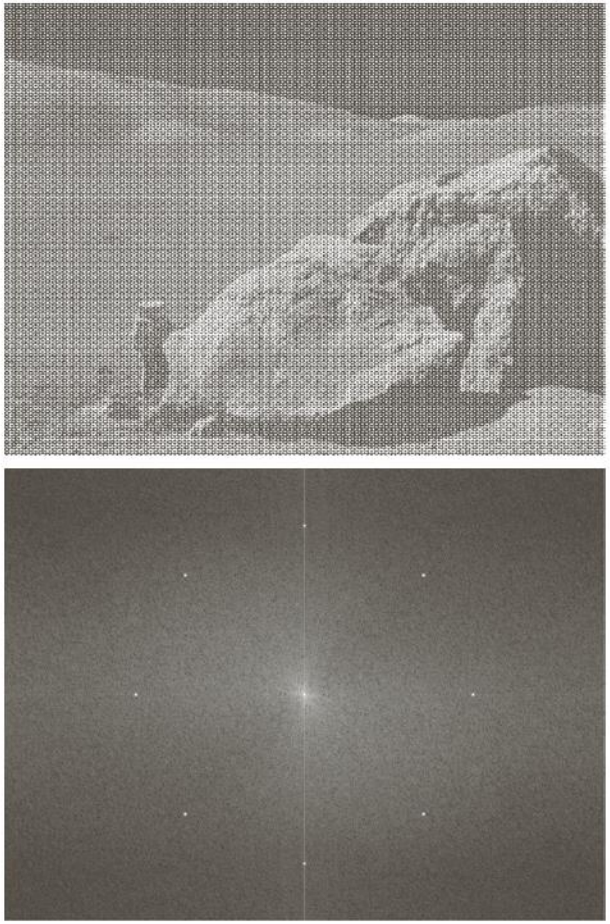


FIGURE 5.4 *(Continued)* Images and histograms resulting from adding exponential, uniform, and salt and pepper noise to the image in Fig. 5.3.



a
b

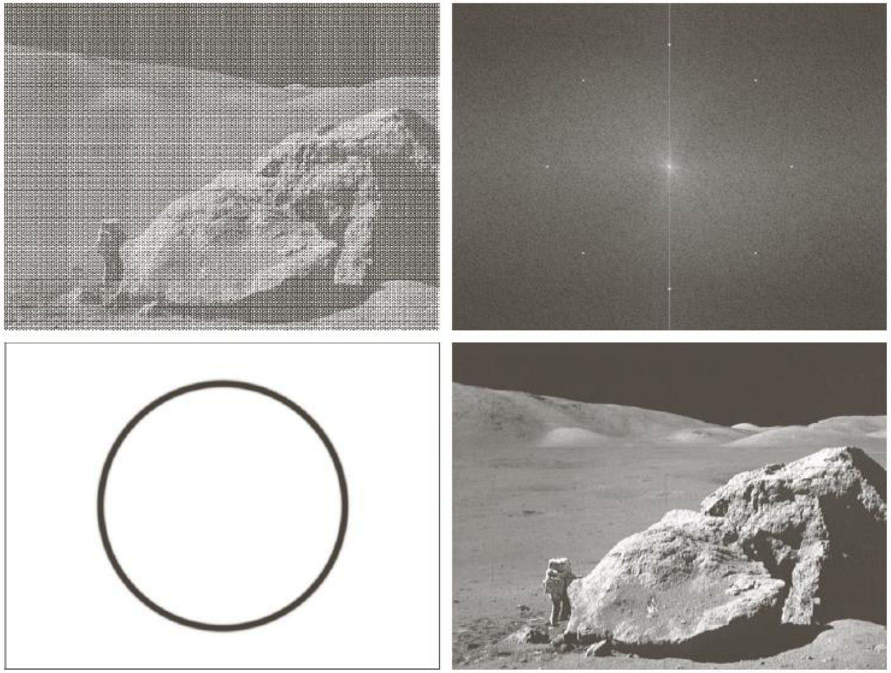
FIGURE 5.5 (a) Image corrupted by sinusoidal noise. (b) Spectrum (each pair of conjugate impulses corresponds to one sine wave). (Original image courtesy of NASA.)

Spatial Filter

- Averaging Smooth (Blur)
- Gaussian Smooth (Blur)
- Media Filter

Filtering In Frequency Domain

- Band Reject Filter
- Notch Reject Filter

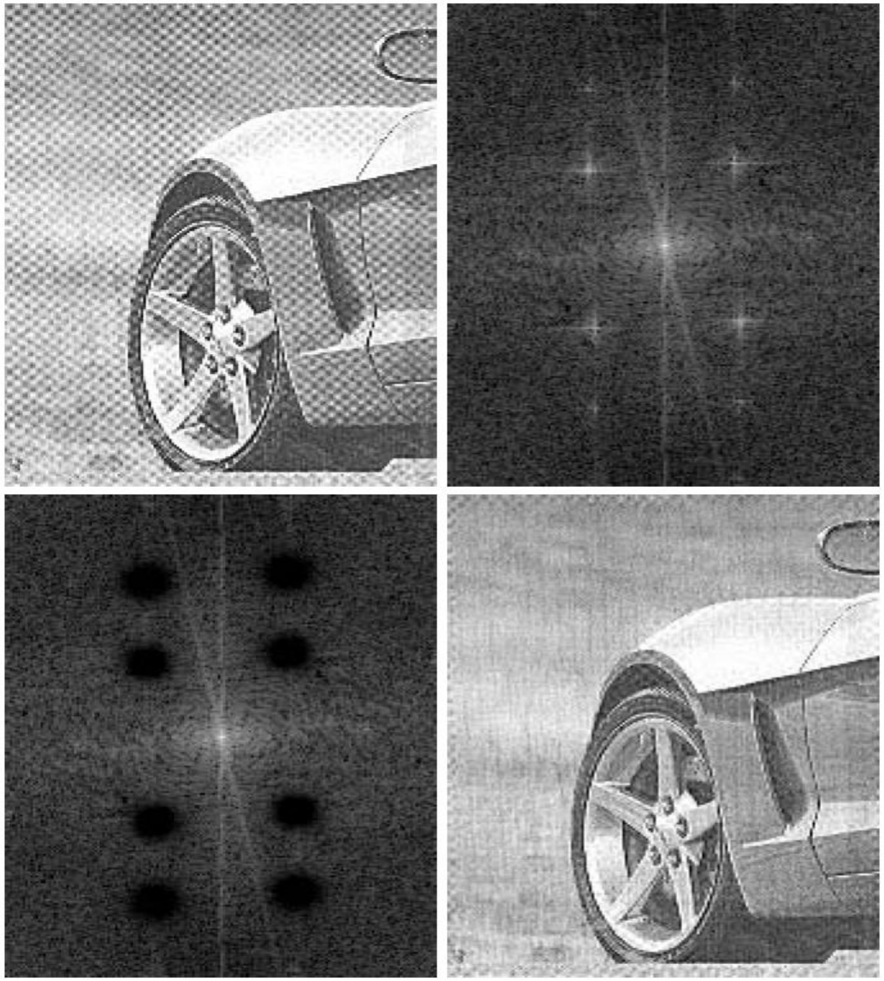


a
b

FIGURE 5.5 (a) Image corrupted by sinusoidal noise. (b) Spectrum (each pair of conjugate impulses corresponds to one sine wave). (Original image courtesy of NASA.)

Filtering In Frequency Domain

- Notch Reject Filter

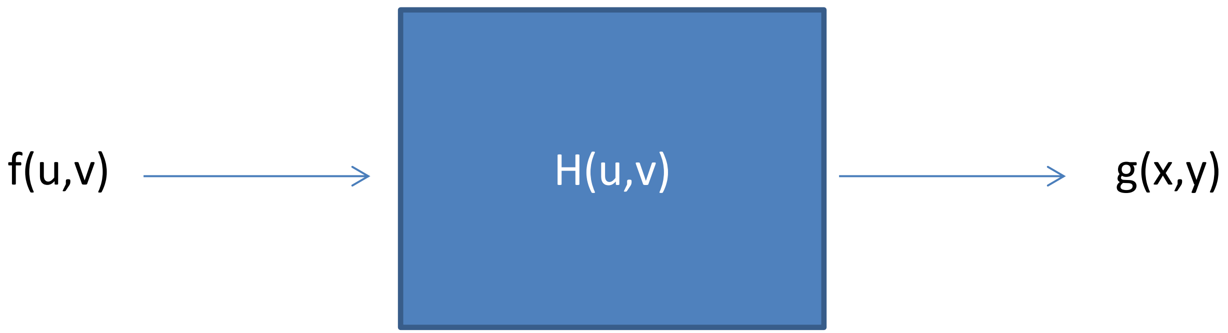


[Programming]

- Generate a Gaussian Band Pass Filter
  - Parameters: r, σ
- Generate a Gaussian Notch Pass Filter
  - Parameters: (u0,v0), σ

Inverse Filtering

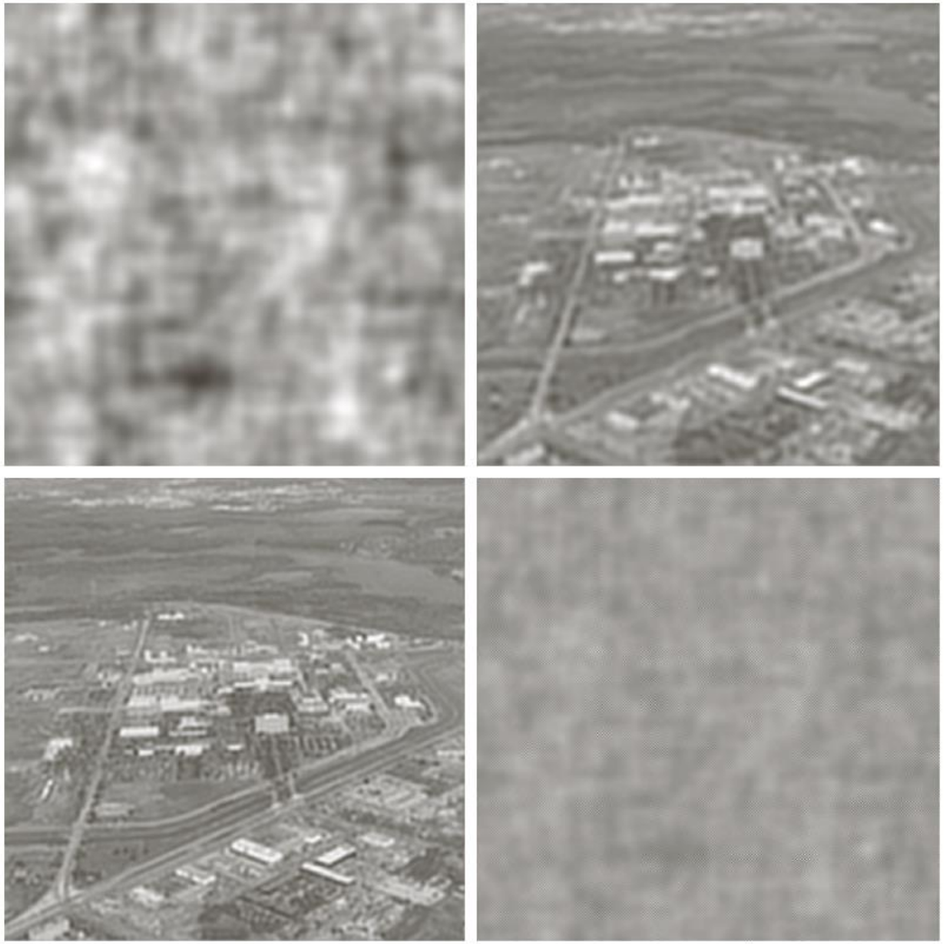
- Linear Degradation: $G(u,v) = F(u,v)\ H(u,v)$



Inverse Filtering

- ?

$$\hat{F}(u,v) = \frac{G(u,v)}{H(u,v)}$$



a b
c d

FIGURE 5.27 Restoring Fig. 5.25(b) with Eq. (5.7-1). (a) Result of using the full filter. (b) Result with H cut off outside a radius of 40; (c) outside a radius of 70; and (d) outside a radius of 85.

a	b
c	d

FIGURE 5.25 Illustration of the atmospheric turbulence model. (a) Negligible turbulence. (b) Severe turbulence, $k = 0.0025$. (c) Mild turbulence, $k = 0.001$. (d) Low turbulence, $k = 0.00025$. (Original image courtesy of NASA.)